Capturing Data Absent Reason for Clinical Concepts Using FHIR R4

A comprehensive guide to documenting missing clinical data — from Allergies and Family History to Procedures and Observations — using the HL7 FHIR R4 standard.

Why Capture Data Absent Reason?

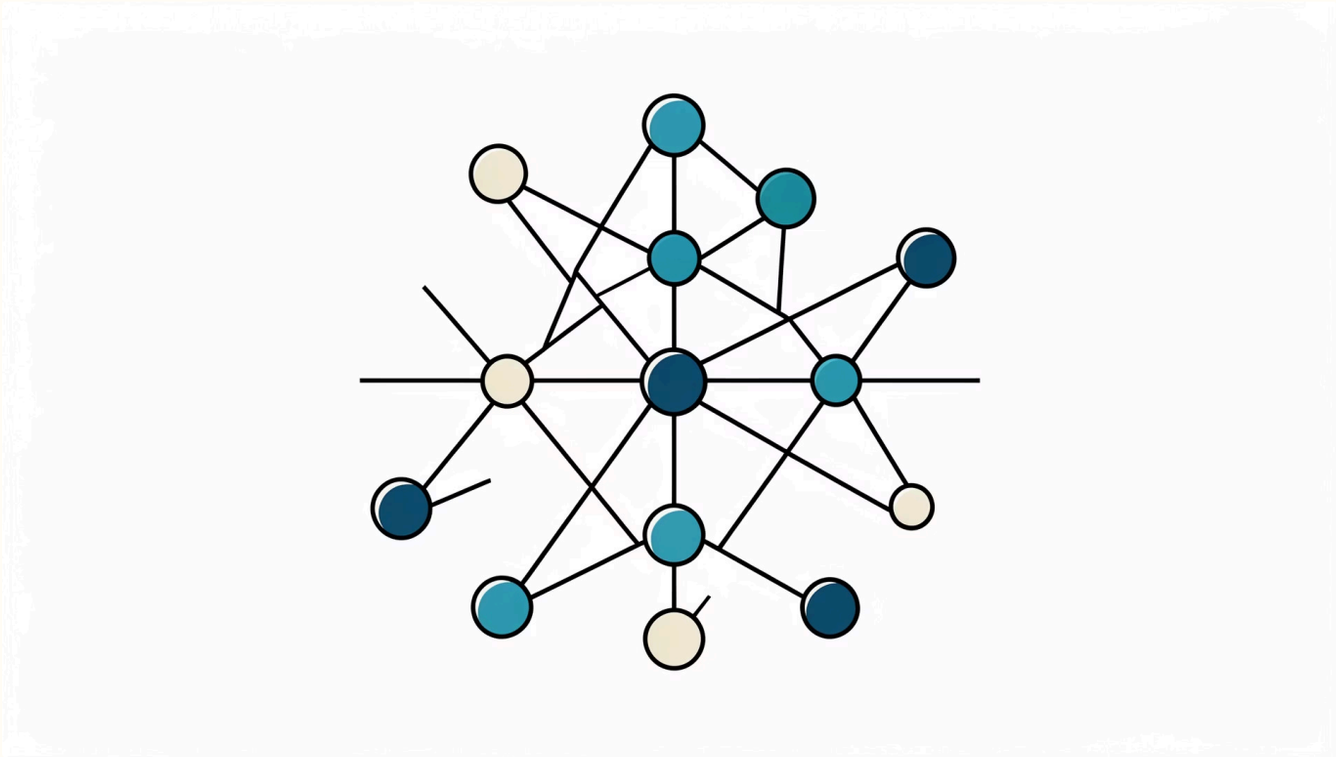
Clinical data is rarely complete. Allergy records, family histories, and procedure details are frequently missing — not because they don't matter, but because the information was **unavailable, not asked, or declined** at the point of care.

Without a standardized way to explain *why* data is absent, downstream systems are left guessing. This ambiguity can compromise:

- **Patient safety** — clinicians may assume "no data" means "no issue"
- **Clinical decision support** — algorithms can't distinguish unknown from negative
- **Interoperability** — receiving systems lose critical context during data exchange

FHIR R4 solves this with a standardized extension that makes absence **explicit, coded, and computable**.

What is Data Absent Reason in FHIR?



The **Data Absent Reason** is an official HL7 FHIR extension that can be applied to *any* element where expected data is missing.

Extension URL:
http://hl7.org/fhir/StructureDefinition/data-absent-reason

It uses a **predefined code set** (the DataAbsentReason ValueSet) to provide a consistent, computable explanation for why data is missing. This enables systems to clearly distinguish between scenarios like:

- The value is **unknown** to the source
- The question was **never asked**
- The patient **declined** to answer
- The data was **masked** for privacy reasons

Key DataAbsentReason Codes

The following codes from the HL7 DataAbsentReason ValueSet cover the most common clinical scenarios for missing data:

Code	Display	When to Use
`unknown`	Unknown	Value expected but not known by anyone
`asked-unknown`	Asked But Unknown	Source was asked but does not know the answer
`not-asked`	Not Asked	Data was never collected; workflow skipped it
`asked-declined`	Asked But Declined	Patient was asked but declined to provide an answer
`masked`	Masked	Data withheld for privacy or security reasons
`not-applicable`	Not Applicable	No proper value exists in this context
`error`	Error	System or workflow error caused data loss

CHAPTER 4 CLINICAL APPLICATION

Applying Data Absent Reason to Clinical Concepts

AllergyIntolerance

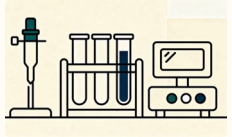
Attach `dataAbsentReason` to **code** or **reaction** elements when allergy substance or severity is unknown or not collected.

FamilyMemberHistory

Apply the extension on **condition** or **relationship** details when family medical history is unavailable or the patient declines to share.

Procedure

Use `dataAbsentReason` on **Procedure.code** or **Procedure.performed** when procedure details or timing are missing from the record.



Observation

FHIR R4 provides a **native** `Observation.dataAbsentReason` element — no extension needed — to explain why a result value is absent.

How to Implement Data Absent Reason in FHIR R4

Identify Missing Element

Locate the element lacking a value.

Apply ValueSet Code

Use appropriate DataAbsentReason code.

Add dataAbsentReason

Attach extension with correct URL.

Example: Missing Allergy Substance

```
{
  "resourceType":
"AllergyIntolerance",
  "code": {
    "extension": [{
      "url":
"http://hl7.org/fhir/StructureDefinition/data-absent-reason",
      "valueCode": "unknown"
    }]
  },
  "patient": {
    "reference": "Patient/123"
  }
}
```

The extension replaces the missing value on the target element, providing a **coded, computable explanation** that receiving systems can interpret programmatically.

Epic Implementation: Allergies & Clinical Lists

Epic captures absent clinical information using **FHIR List resources** — a pattern where an empty list with a coded `emptyReason` communicates that data was reviewed but intentionally absent.

1

List-Based Pattern

Epic uses `List` resources for Allergies, Problems, and Medications. An empty list with `emptyReason` (e.g., `nilknown`) signals "reviewed, none known."

2

Negation Coding

For allergies, Epic may return a coded entry like **"No Known Allergies"** (SNOMED `716186003`) rather than using the extension directly on individual elements.

3

Interoperability Consideration

Consumers should check both the `List.emptyReason` and individual resource extensions when integrating with Epic's FHIR APIs to capture the full absent-data picture.

Cerner Implementation of Data Absent Reason

Oracle Health (Cerner) leverages the **dataAbsentReason extension** directly on FHIR resource elements, aligning closely with the HL7 specification.

Extension-First Approach

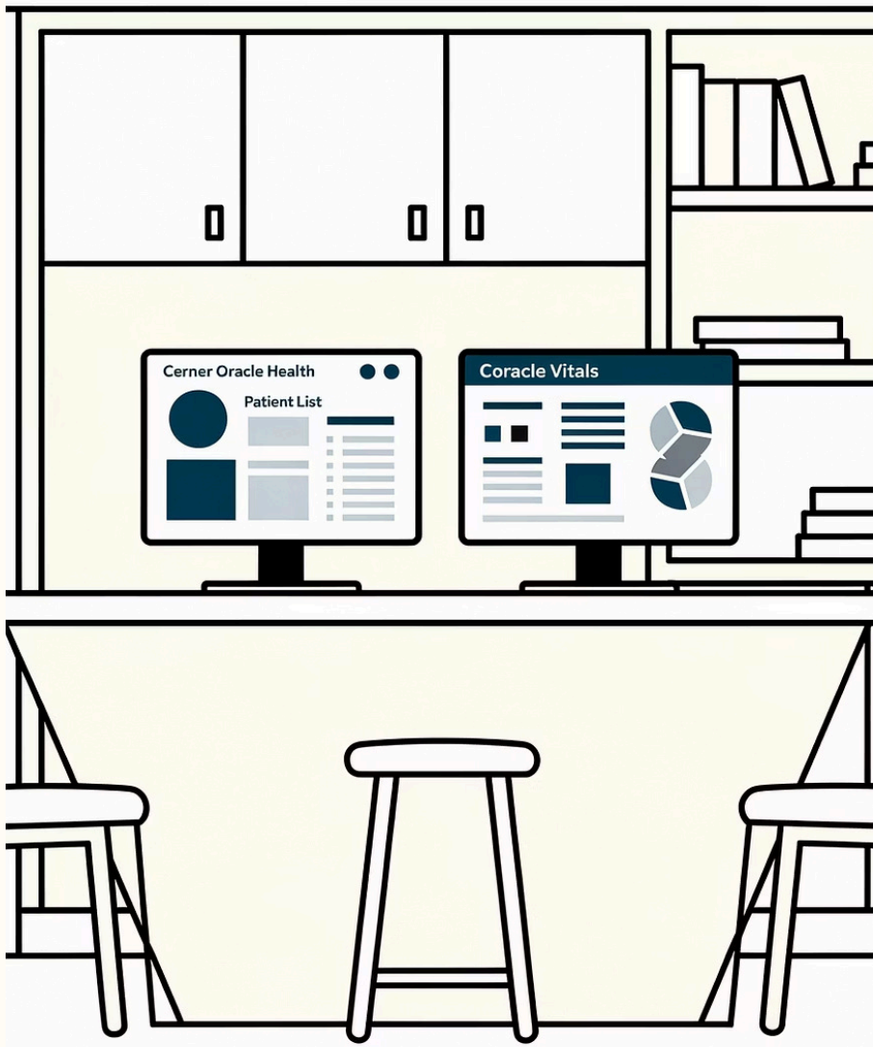
Cerner applies the standard data-absent-reason extension on elements like **Observation.value** and **AllergyIntolerance.code** when data is missing.

Common Codes Used

Frequently returned codes include unknown, not-asked, and masked — reflecting real clinical workflows and privacy policies.

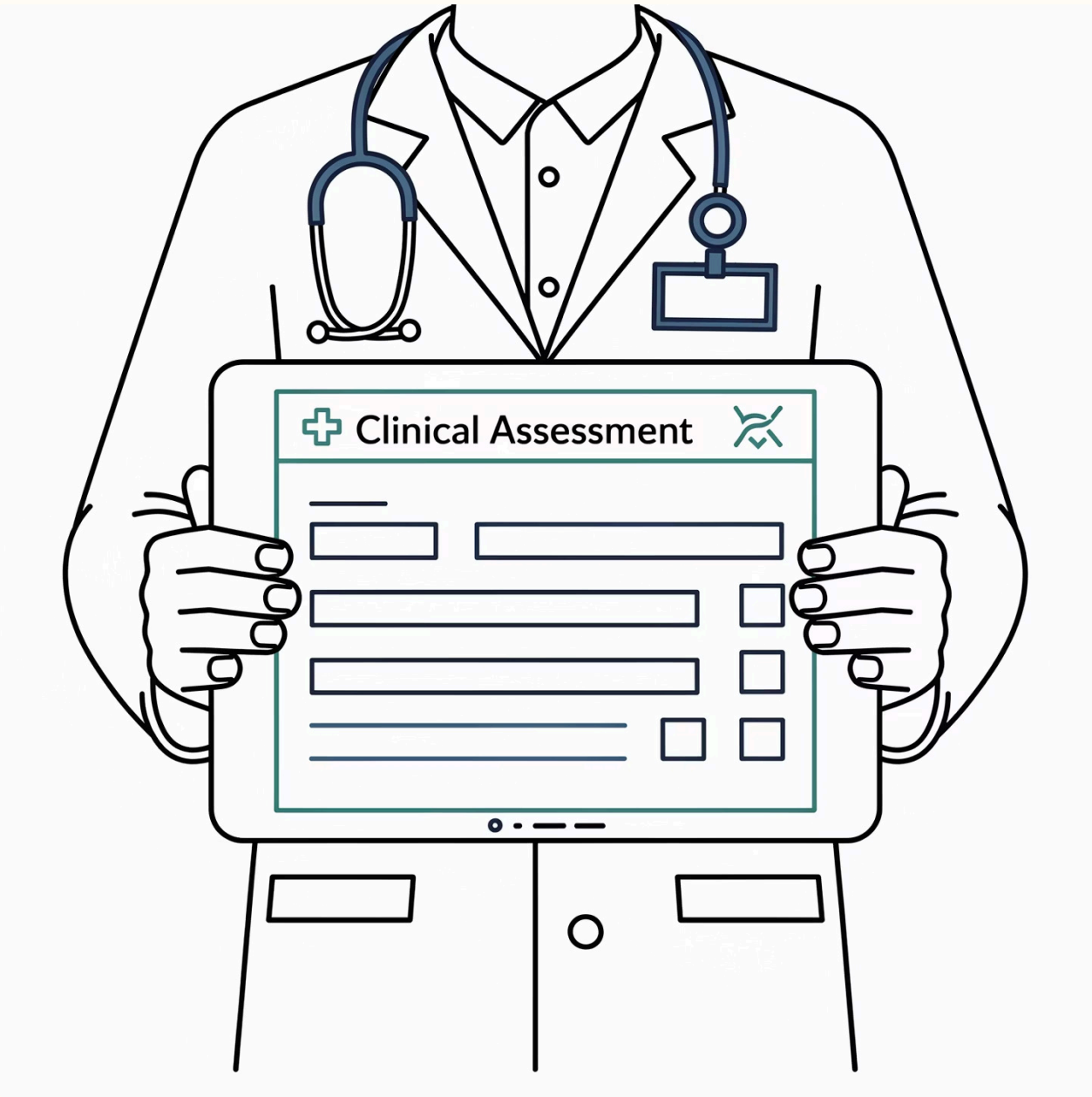
API Behavior

Cerner's FHIR R4 APIs surface absent reasons in search results, enabling consuming applications to render appropriate UI indicators for missing data.



Structured Data Capture (SDC) & Data Absent Reason

Structured Data Capture (SDC) is an HL7 FHIR implementation guide that standardizes how questionnaires and assessments are defined and completed using the Questionnaire and QuestionnaireResponse resources. When respondents cannot or do not answer a question, Data Absent Reason provides a computable way to explain why.



SDC IG URL: http://hl7.org/fhir/uv/sdc/

- SDC uses FHIR Questionnaire to define assessment structure
- QuestionnaireResponse captures answers (or their absence)
- Data Absent Reason extension applies to QuestionnaireResponse.item.answer when a response is missing
- Enables downstream analytics to distinguish "not answered" from "answered negatively"

Data Absent Reason in QuestionnaireResponse

When a patient or clinician cannot answer a question in a Structured Data Capture (SDC) assessment, the `Data Absent Reason` extension is applied directly to the `answer` element of a `QuestionnaireResponse` item.

Example: Patient Declines to Answer

```
{
  "resourceType": "QuestionnaireResponse",
  "item": [{
    "linkId": "1.1",
    "text": "Do you have a family history of heart disease?",
    "answer": [{
      "extension": [{
        "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason",
        "valueCode": "asked-declined"
      }]
    }]
  }]
}
```

This example demonstrates how the extension provides a computable explanation for the missing answer, indicating the patient chose not to provide the information.

`asked-declined`

Patient declined to answer a sensitive assessment question (e.g., mental health, substance use).

`not-asked`

Question was skipped due to conditional logic or workflow branching.

`unknown`

Respondent was unable to recall or provide the requested information.

`masked`

Answer exists but is withheld for privacy or role-based access reasons.

Conditional Logic, Prepopulation & Absent Reason

Structured Data Capture (SDC) goes beyond simple questionnaires, incorporating advanced logic and data flows. Understanding how Data Absent Reason (DAR) interacts with these complex SDC features is crucial for maintaining data integrity throughout the clinical workflow.

EnableWhen & Skipped Questions

When SDC `enableWhen` conditions cause a question to be hidden or skipped, the resulting `QuestionnaireResponse` item may carry a `not-asked` `dataAbsentReason`. This signals that the question was intentionally bypassed by workflow logic, not merely forgotten or ignored by the respondent.

Prepopulation from EHR Data

SDC supports prepopulating answers from existing FHIR resources (e.g., pulling a patient's known allergies or medical history). If the source data itself carries a `dataAbsentReason`, that reason should be accurately propagated into the `QuestionnaireResponse` answer to preserve the clinical context and data provenance.

Extraction & Downstream Use

When SDC responses are extracted back into canonical FHIR resources (such as `Observation`, `Condition`, or `MedicationStatement`), any `dataAbsentReason` on a `QuestionnaireResponse` answer must be carefully mapped to the corresponding element on the target resource. This ensures the absent-data signal is not lost during data transformation and remains computable for downstream systems.

Best Practice: Always propagate Data Absent Reason codes through the full SDC lifecycle — from source data, through the QuestionnaireResponse, to extracted FHIR resources — to maintain a complete and trustworthy clinical record.